

Multi-Task Graph Neural Networks for Political Account Identification and Misinformation Label Prediction

Political Misinformation Social Network Project

Abstract

This paper investigates multi-task graph neural networks for the analysis of political communication on social media. A collection of 7,053 tweets from 14 Macedonian political accounts is represented as a graph containing 10,160 nodes and 97,364 edges. A shared graph encoder is trained with two task-specific prediction heads: a 14-class author-identification head and a binary classifier for automatically generated misinformation labels. Graph Convolutional Networks, GraphSAGE, and Graph Attention Networks are evaluated under a common stratified data split. GraphSAGE provides the strongest balance between the two tasks, obtaining an author macro-F1 of 0.989 and a misinformation macro-F1 of 0.708. The findings demonstrate the usefulness of graph representation learning for account identification while also highlighting the limitations imposed by weakly supervised misinformation labels.

Keywords: graph neural networks; social networks; political communication; misinformation; multi-task learning

1. Introduction

Political discussion on social media is shaped by both the content of individual messages and the network structure through which messages are produced and circulated. Treating tweets as independent observations discards information about authorship, interaction, and shared social context. Graph neural networks provide a natural framework for combining these sources of information by propagating feature representations over a graph. This study examines a multi-task formulation of political tweet analysis. The first task is author identification, while the second predicts whether a tweet received a generated misinformation label. The tasks share a graph encoder but use separate output heads.

2. Related Work

Graph convolutional networks perform localized neighborhood aggregation for semi-supervised node classification. GraphSAGE learns an aggregation function that can be applied to previously unseen nodes, while graph attention networks use learned attention coefficients to weight neighboring representations. This work applies these architectures to political social-network data and explicitly treats the misinformation task as weak supervision because its labels are generated by a zero-shot language model rather than an independent fact-checking process.

3. Data and Graph Representation

The working archive contains 7,053 tweets associated with 14 political accounts. Each tweet is represented using text-derived and metadata features prepared by the original preprocessing workflow. Social-network entities and relations extracted from the archive are added to form a graph represented in PyTorch Geometric. The resulting graph contains 10,160 nodes and 97,364 edges. Tweet nodes occur first in the node-feature matrix and are the only nodes used as labeled examples; the remaining nodes provide neighborhood context during message passing. Misinformation labels were generated using joeddav/xlm-roberta-large-xnli with the candidate labels valid, misleading, and invalid. The resulting class distribution is 5,303 valid, 1,084 invalid, and 666 misleading tweets. These are pseudo-labels rather than expert fact-checks.

4. Methodology

Each model consists of two graph convolution layers with a hidden dimension of 96. Each layer is followed by layer normalization, a ReLU nonlinearity, and dropout. The shared representation feeds a 14-class author head and a binary misinformation head. The objective is the sum of the two cross-entropy losses, with the

positive misinformation class weighted by 2.0. The comparison includes GCN, GraphSAGE, and GAT; all other components are held constant. Tweet nodes are divided into training, validation, and test partitions using an 80/20 test split followed by an 80/20 training-validation split. Stratification is performed using author labels, and the same partition is reused for all architectures. Accuracy and macro-F1 are reported for each task.

5. Results

Model	Author accuracy	Author macro-F1	Misinformation accuracy	Misinformation macro-F1
GCN	0.962	0.957	0.755	0.689
GraphSAGE	0.991	0.989	0.775	0.708
GAT	0.997	0.997	0.703	0.641

6. Discussion

All three architectures perform strongly on author identification. GAT obtains the highest author score, followed closely by GraphSAGE. The misinformation task is more difficult: GraphSAGE obtains the highest misinformation accuracy and macro-F1, while GAT performs worst on this task despite its superior author results. The divergence indicates that account-specific graph structure is highly informative, but does not transfer equally well to the noisier misinformation labels. These results do not establish that the models can determine factual truth; they measure agreement with generated labels.

7. Limitations and Future Work

The principal limitation is the absence of independently verified misinformation annotations. Future work should replace generated labels with fact-checking labels or annotate a carefully sampled evaluation set. A temporal split would provide a more realistic estimate of generalization to later political events. Further experiments could compare heterogeneous graph formulations, tune task-loss weights, and report calibration and per-class confusion matrices.

8. Conclusion

This paper presented a multi-task GNN framework for political tweet analysis. GraphSAGE achieved the best balance between political-account identification and prediction of generated misinformation labels, while GAT achieved the strongest author-identification results. The study demonstrates the practical value of graph representation learning for social-media analysis while emphasizing that weakly supervised misinformation prediction must be interpreted cautiously.

Reproducibility

The preprocessing notebooks, model implementation, dependency list, and Google Colab workflow are available at github.com/VikiShmiki/Political-Misinformation-Social-Networks.

References

- Kipf, T. N., and Welling, M. (2017). Semi-Supervised Classification with Graph Convolutional Networks. ICLR.
- Hamilton, W. L., Ying, R., and Leskovec, J. (2017). Inductive Representation Learning on Large Graphs. NeurIPS.
- Velickovic, P. et al. (2018). Graph Attention Networks. ICLR.